

TOWERS OF HANOI

AIM

To write a Prolog program to implement the Towers of Hanoi problem using recursion.

OBJECTIVE

- To understand recursion in Prolog.
- To solve the Towers of Hanoi problem.
- To move N disks from a source peg to a destination peg using an auxiliary peg.
- To follow the rule that a larger disk cannot be placed on a smaller disk.

ALGORITHM

1. Start.
2. If there is only one disk, move it directly from source to destination.
3. For N disks:
 - Move N-1 disks from source to auxiliary.
 - Move the largest disk from source to destination.
 - Move N-1 disks from auxiliary to destination.
4. Repeat recursively.
5. Display all moves.
6. Stop.

SOURCE CODE

% Towers of Hanoi

hanoi(1, Source, Destination, _) :-

```
    write('Move disk from '),  
    write(Source),  
    write(' to '),  
    write(Destination),  
    nl.
```

hanoi(N, Source, Destination, Auxiliary) :-

N > 1,

N1 is N - 1,

hanoi(N1, Source, Auxiliary, Destination),

write('Move disk from '),

write(Source),

write(' to '),

write(Destination),

nl,

hanoi(N1, Auxiliary, Destination, Source).

OUTPUT

```
SWI-Prolog: no-true program in debug mode (.- debug.) to get more details.  
?- move(3,left,right,center).  
Move the top disk from left to right  
Move the top disk from left to center  
Move the top disk from right to center  
Move the top disk from left to right  
Move the top disk from center to left  
Move the top disk from center to right  
Move the top disk from left to right  
true ■
```

RESULT

The Prolog program successfully solves the Towers of Hanoi problem using recursion. For 3 disks, it performs 7 moves, which is the minimum required ($2^3 - 1$).